

Introduction

We hope to solve the equation

$$i\partial_t\psi = (-i\sigma_x\partial_x + m\sigma_z + f(x))\psi$$

where $\psi = (\psi_1\psi_2)^T$, and f is some potential function. In theory, this is only slightly different from Schrödinger. If we can store the spinor ψ , nearly all the same methods can be used to solve Dirac; we will use Strang splitting to approximate the solution by solving each term individually.

Discretization and Storage

Fortunately, the problem of storing the spinor on qubits has a simple solution: we maintain a register to store position, as in Schrödinger, and add one qubit as an indicator. We discretize ψ_1 and ψ_2 into $|\psi_1\rangle$ and $|\psi_2\rangle$ as we know how from Schrödinger, and store the state ψ as

$$\begin{aligned} |\psi\rangle &\propto \sum_{k=0}^{N-1} (\psi_1(x_k) |0\rangle + \psi_2(x_k) |1\rangle) \otimes |k\rangle \\ &= |0\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_1(x_k) |k\rangle \right) + |1\rangle \otimes \left(\sum_{k=0}^{N-1} \psi_2(x_k) |k\rangle \right) \\ &\propto |0\rangle \otimes |\psi_1\rangle + |1\rangle \otimes |\psi_2\rangle \end{aligned}$$

With the indicator as the most significant qubit, this is equivalent to concatenating the column vectors of amplitudes for $|\psi_1\rangle$ and $|\psi_2\rangle$.

Transport Term